

Q1.

```
class Node {  
  
    Node next;  
    int data;  
  
    Node(int data){  
        this.data = data;  
    }  
  
    public void printDetails(){  
        System.out.println(data+" ");  
    }  
}
```

```
class LinkedList {  
  
    Node head;  
    Node tail;  
  
    LinkedList() {  
        this.head = null;  
        this.tail = null;  
    }  
  
    public void insert(int new_data) {  
        Node new_node = new Node(new_data);  
  
        if (head == null) {  
            head = new_node;  
            tail = new_node;  
        } else {  
            tail.next = new_node;  
            tail = new_node;  
        }  
    }  
  
    public boolean nodeExists(Node head, int x) {  
        Node current = head;  
        while (current != null) {  
            if (current.data == x) {  
                return true;  
            }  
        }  
    }  
}
```

```

    }
    current = current.next;
}
return false;
}

public Node search(int x) {
    Node current = head;
    if (nodeExists(current, x) == true) {
        while (current != null) {
            if (current.data == x) {
                return current;
            }
            current = current.next;
        }

    } else {
        return null;
    }
    return null;
}

```

```

void remove(int data) {

    Node temp = head, prev = null;
    if (temp == search(data)) {
        head = temp.next;
        return;
    }
    while (temp != search(data)) {
        prev = temp;
        temp = temp.next;
    }
    if (temp == null) {
        return;
    }
    prev.next = temp.next;
}

```

```

void display() {
    Node temp = head;
    while (temp != null) {
        System.out.print(temp.data + " ");
        temp = temp.next;
    }
}

```

```
}
```

```
class HashTable{
    private static final int SIZE_OF_ARRAY = 5;
    LinkedList indexes[] = new LinkedList[SIZE_OF_ARRAY];

    public int hashCode(int data){

        return data%SIZE_OF_ARRAY;
    }

    public void insert(int data){
        int index = hashCode(data);

        if(indexes[index]==null){
            LinkedList newList = new LinkedList();
            indexes[index] = newList;
            newList.insert(data);

        }else{
            LinkedList temp = indexes[index];
            temp.insert(data);

        }

    }

    public void retrieve(int data){

        int index = hashCode(data);
        if(indexes[index]==null){
            System.out.println("Item Not Found");
        }else{
            indexes[index].remove(data);
        }

    }

    public void Display(){

        for(int i=0; i<SIZE_OF_ARRAY;i++){

            LinkedList temp = indexes[i];
            if(temp!=null){
                temp.display();
                System.out.println("");
            }

        }

    }

}
```

```
}  
}
```

```
}
```

```
}
```

```
class Driver {
```

```
    public static void main(String args[]){
```

```
        HashTable hashtable = new HashTable();
```

```
        hashtable.insert(5);  
        hashtable.insert(10);  
        hashtable.insert(20);  
        hashtable.insert(1);  
        hashtable.insert(0);  
        hashtable.insert(100);  
        hashtable.insert(13);  
        hashtable.insert(21);
```

```
        hashtable.Display();
```

```
        hashtable.retrive(100);
```

```
        System.out.println("after deletion");  
        hashtable.Display();
```

```
    }
```

```
}
```

Q2

```
public class Node {
```

```
    Node next;
```

```
    Book data;
```

```
    public Node(Book data) {
```

```
        this.next = null;
```

```
        this.data = data;
```

```
    }
```

```
    public void printDetails() {
```

```
        System.out.println(data + " ");
```

```
    }
```

```
}
```

```
// Book class
```

```
public class Book {
```

```
    private String book_id;
```

```
    private String book_name;
```

```
    private String author;
```

```
    private int num_pages;
```

```
    private int ratings;
```

```
    public Book(String id, String name, String author, int pages, int ratings) {
```

```
        this.book_id = id;
```

```
        this.book_name = name;
```

```
        this.author = author;
```

```
        this.num_pages = pages;
```

```
        this.ratings = ratings;
```

```
    }
```

```
    public String getBook_id() {
```

```
        return book_id;
```

```
    }
```

```
    public String getAuthor() {
```

```
        return author;
```

```

    }

    public String getBook_name() {
        return book_name;
    }

    public int getNum_pages() {
        return num_pages;
    }

    public int getRatings() {
        return ratings;
    }
}

```

//Linked list

```

public class LinkedList {

    Node head;
    Node tail;

    public LinkedList() {
        this.head = null;
        this.tail = null;
    }

    public void insertRear(Book book) {
        Node new_node = new Node(book);

        if (head == null) {
            head = new_node;
            tail = new_node;
        } else {
            tail.next = new_node;
            tail = new_node;
        }
    }

    public Node searchLinkedList(String author) {

        Node temp = head, prev = null;
        if (temp.data.getAuthor().equals(author)) {
            this.head = head.next;
            return temp;
        }
    }
}

```

```

while (temp.data.getAuthor().equals(author) != true) {
    prev = temp;
    temp = temp.next;
}

prev.next = temp.next;
return temp;

}

public void displayList() {
    Node temp = head;
    while (temp != null) {

        System.out.println(
            "\t" + temp.data.getBook_id() + "," + temp.data.getBook_name() + "," +
temp.data.getAuthor() + ","
            + temp.data.getNum_pages() + "," + temp.data.getRatings());

        temp = temp.next;
    }

}
}

```

//HashTable

```

public class HashTable {
    private static final int SIZE_OF_ARRAY = 10;
    private LinkedList indexes[] = new LinkedList[SIZE_OF_ARRAY];

    public int hashCode(String author) {
        int ascii = 0;
        for (int i = 0; i < author.length(); i++) {

            ascii = author.charAt(i) + ascii;

        }
        return ascii % SIZE_OF_ARRAY;
    }

    public void Insert(Book book) {
        int index = hashCode(book.getAuthor());

        if (indexes[index] == null) {

```

```
        LinkedList list = new LinkedList();
        indexes[index] = list;
        list.insertRear(book);
    } else {
        LinkedList temp = indexes[index];
        temp.insertRear(book);
    }
}

public Book retrieve(String author) {
    int index = hashCode(author);
    return indexes[index].searchLinkedList(author).data;
}

public void Display() {
    for (int i = 0; i < SIZE_OF_ARRAY; i++) {
        if (indexes[i] != null) {

            indexes[i].displayList();
            System.out.println("");
        }
    }
}
}
```

```

package Q2;

public class Driver {

    public static void main(String[] args) {

        HashTable hashtable = new HashTable();
        hashtable.Insert(new Book("B01", "Harry Potter", "Rowling", 652, 2095690));
        hashtable.Insert(new Book("B02", "The Lost Continent", "Bryson", 299, 45712));
        hashtable.Insert(new Book("B03", "The Lord of the Rings", "Tolkein", 1184, 1710));
        hashtable.Insert(new Book("B04", "Notes from a Small Island", "Bryson", 324, 80609));
        hashtable.Insert(new Book("B05", "The Changeling Sea ", "Patricia", 137, 4454));
        hashtable.Insert(new Book("B06", "Heirs of General Practice", "McPhee", 128, 268));
        hashtable.Insert(new Book("B07", "Salmon of Doubt", "Douglas", 336, 5));
        hashtable.Insert(new Book("B08", "For the New Intellectual", "Rand", 224, 2750));
        hashtable.Insert(new Book("B09", "City of Glass", "Auster", 203, 12410));
        hashtable.Insert(new Book("B10", "A War Like No Other", "Davis", 397, 1693));

        hashtable.Display();

        hashtable.retrieve("Tolkein");
        System.out.println("*****");

        hashtable.Display();

    }
}

```